

Java Full Stack Development

1. Scanning



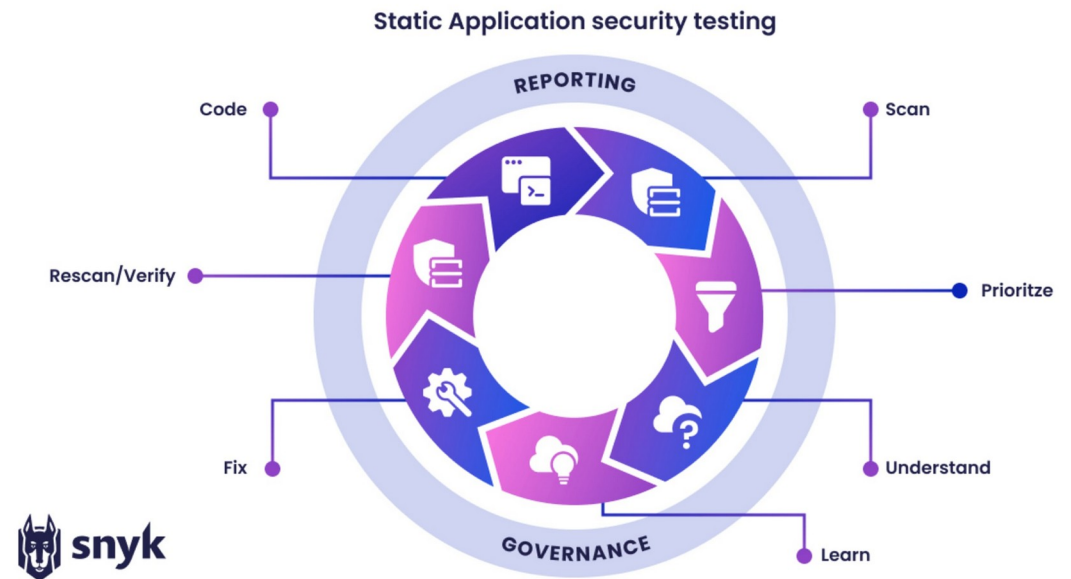
Module Topics

- Static Security Application Testing
- Dynamic Security Application Testing

Static Security Application Testing

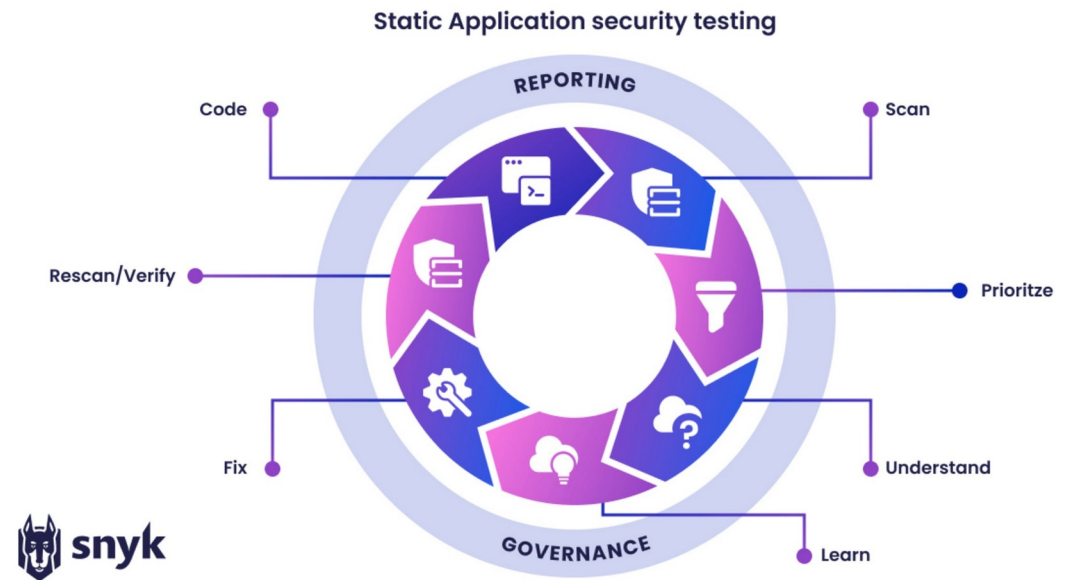
SAST

- Static Application Security Testing
 - Static
 - Analyzes source code or compiled bytecode without executing it
 - Application
 - Focuses on the code your team writes, not the OS or infrastructure
 - Security Testing
 - Looks specifically for security defects, not bugs in general
 - Runs early and often
 - Every commit, every pull request, every CI build
 - The "shift left" philosophy
 - Find vulnerabilities at the cheapest point in the SDLC



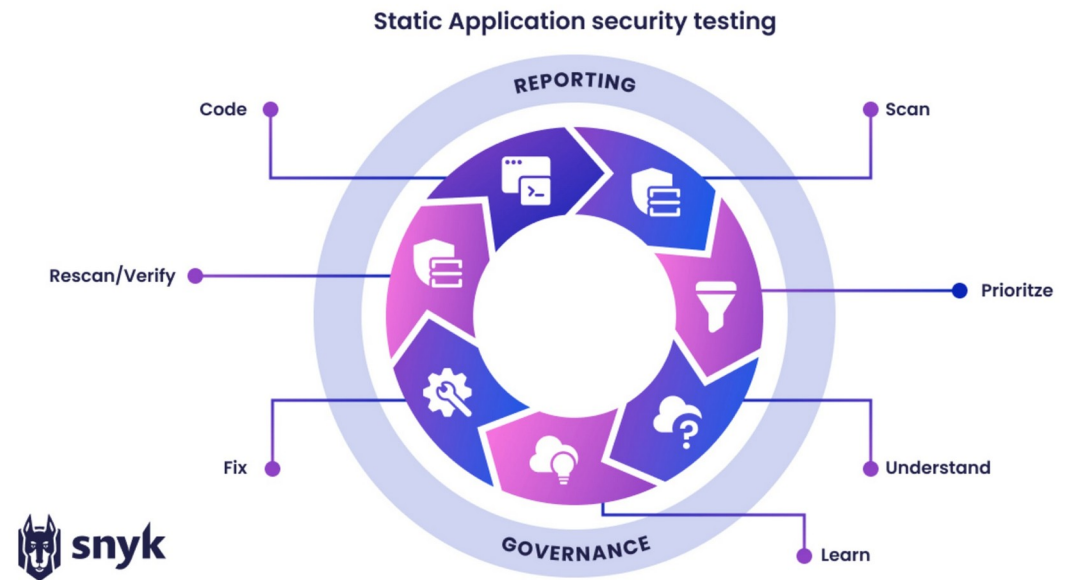
SAST

- Static
 - Code is analyzed without being runC
 - The analyzer reads the source files
 - Builds a model of what the code does
 - Looks for patterns that indicate vulnerabilities
 - Compare with dynamic analysis (DAST), which runs the application and probes it from outside
 - Static is faster, cheaper, and earlier in the SDLC but dynamic catches things static cannot
- Application
 - For application code
 - Not for operating system vulnerabilities, network misconfigurations, or hardware issues
 - Those are covered by other tools



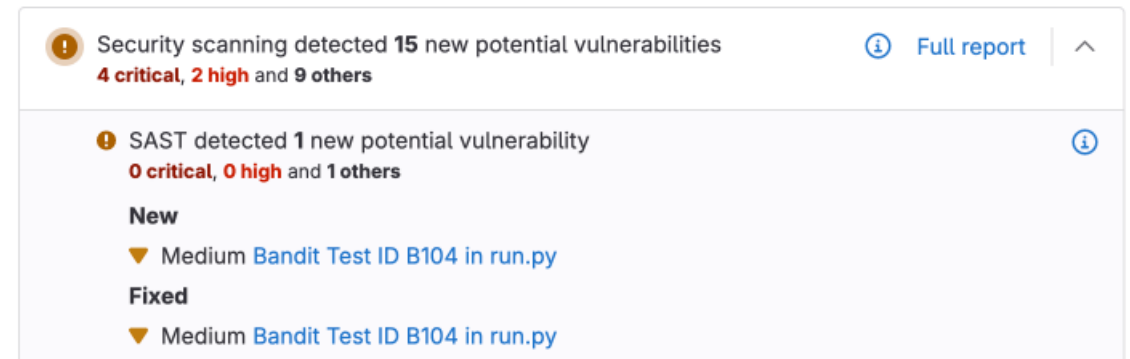
SAST

- Security testing focus
 - SAST tools are tuned to find security defects
 - SQL injection, XSS, hardcoded credentials, weak crypto
 - Some tools also report code quality issues alongside security issues
 - SonarQube is a SAST tool that
 - Does security scans
 - Identifies potential bugs
 - Identifies code smells
 - Provides metrics like cyclomatic complexity
 - Checkmarx is used by enterprises
 - More thorough but requires a licence



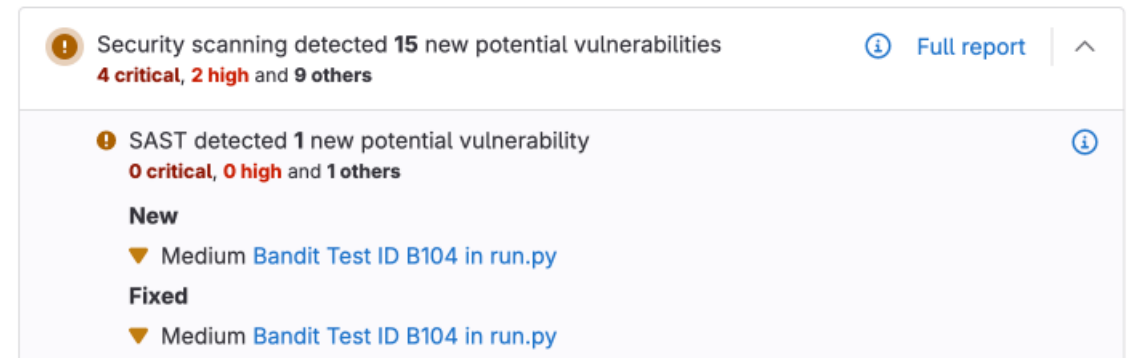
SAST

- How SAST works
 - Step 1: Parse
 - Read source code, build an Abstract Syntax Tree (AST)
 - Step 2: Model
 - Build a control flow graph showing how execution moves through the code
 - Step 3: Analyze
 - Perform taint analysis, tracking untrusted data from sources to sinks
 - Step 4: Report
 - Generate findings for code patterns that match known vulnerability rules



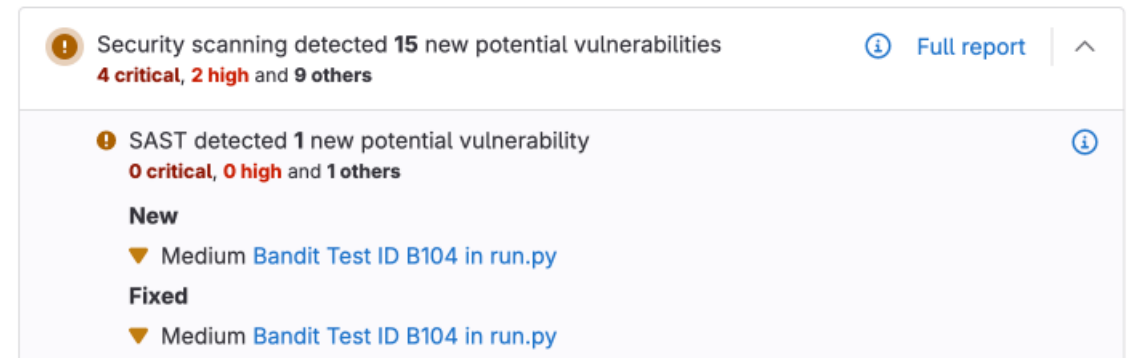
SAST

- What SAST catches
 - Injection vulnerabilities
 - SQL injection, command injection, LDAP injection, XPath injection
 - Cross-site scripting (XSS)
 - Unescaped user input in HTML output
 - Hardcoded credentials
 - Passwords, API keys, tokens in source files
 - Weak cryptography
 - MD5/SHA-1 for security purposes, ECB mode, hardcoded keys, weak random number generation



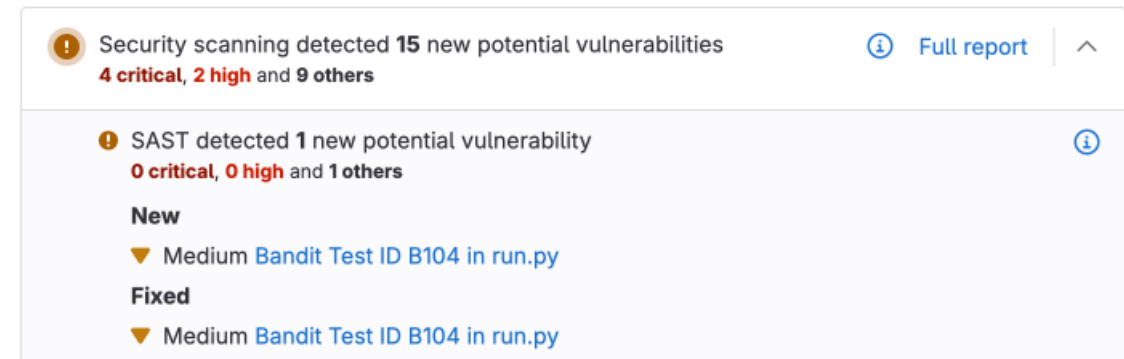
SAST

- What SAST catches
 - Path traversal
 - User input controlling file paths
 - Insecure deserialization
 - Untrusted data passed to readObject(), Jackson, etc.
 - Information leakage
 - Logging sensitive data, exposing stack traces, debug info in responses



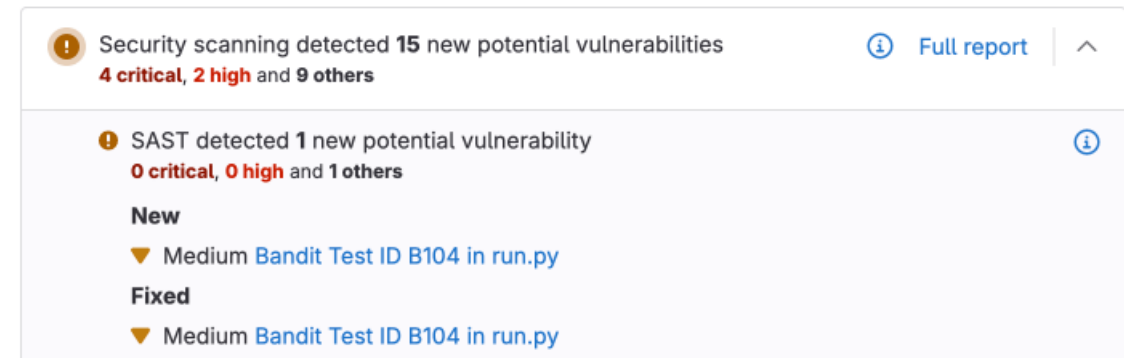
SAST

- Injection
 - The canonical example
 - The pattern is always source-flows-to-sink-without-sanitizer
 - SAST tools detect it well
 - These all share the pattern
 - SQL injection (CWE-89)
 - Command injection (CWE-78)
 - LDAP injection (CWE-90)
 - XPath injection (CWE-91)
 - Header injection (CWE-93)



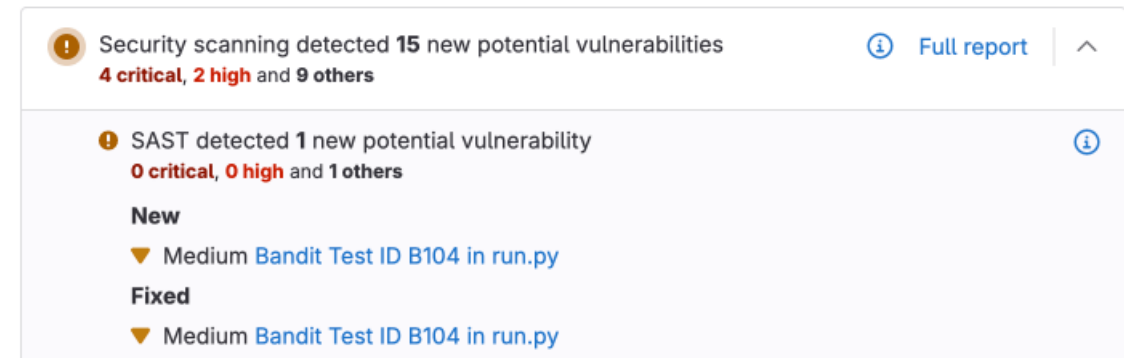
SAST

- Hardcoded credentials
 - Detected by pattern matching
 - Looking for string literals that look like passwords or keys, often with regex against common variable names
 - Less sophisticated than taint analysis but quite effective
- Path traversal
 - Taint analysis applied to file system operations
 - User input flows to new File(...) or Path.of(...)
 - The analyzer asks whether the input was sanitized before use



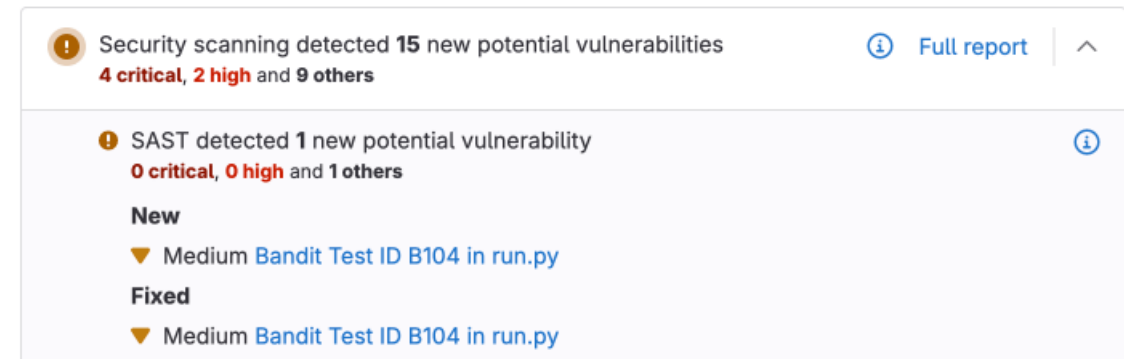
SAST

- What SAST misses
 - Business logic flaws
 - "Users can transfer money from accounts they don't own"
 - Authentication misconfiguration
 - Security rules that aren't applied to the right URLs
 - Authorization bugs
 - Checking the wrong role for the wrong resource
 - Race conditions
 - Concurrency bugs that depend on timing



SAST

- What SAST misses
 - Cryptographic protocol misuse
 - Using crypto correctly per API but wrong for the threat model
 - Information disclosure through observable behavior
 - Timing attacks, error message variation
 - Anything requiring runtime context
 - Environment variables, deployed configuration, user roles in production



CWE

- Common Weakness Enumeration (CWE)

- CWE is a community-maintained taxonomy of software weaknesses
- Every common vulnerability class has a CWE number
 - CWE-89 SQL Injection, CWE-79 XSS, CWE-798 Hardcoded Credentials, CWE-22 Path Traversal
- Every modern SAST tool tags findings with CWE numbers
 - A finding tagged "CWE-89" means the same thing in Checkmarx, SonarQube, Snyk, Semgrep
- The OWASP top 10 maps to CWE categories
 - The Top 10 is a curated list
 - The CWE is the vocabulary

CWE top 25 most dangerous software weaknesses

#1 CWE-787 Out-of-bounds Write	#2 CWE-79 Cross-site Scripting	#3 CWE-89 SQL Injection	#4 CWE-416 Use After Free	#5 CWE-78 OS Command Injection
#6 CWE-20 Improper Input Validation	#7 CWE-125 Out-of-Bounds Read	#8 CWE-22 Path Traversal	#9 CWE-352 Cross-Site Request Forgery	#10 CWE-434 Unrestricted Upload of File
#11 CWE-862 Missing Authorization	#12 CWE-476 NULL Pointer Dereference	#13 CWE-287 Improper Authentication	#14 CWE-190 Integer Overflow or Wraparound	#15 CWE-502 Deserialization of Untrusted Data
#16 CWE-77 Command Injection	#17 CWE-119 Improper Restriction of Operations...	#18 CWE-798 Use of Hard-coded Credentials	#19 CWE-918 Server-Side Request Forgery	#20 CWE-306 Missing Authentication for Critical...
#21 CWE-362 Race Condition	#22 CWE-269 Improper Privilege Management	#23 CWE-94 Code Injection	#24 CWE-863 Incorrect Authorization	#25 CWE-276 Incorrect Default Permissions

CWE

- A few of the most common CWEs
 - CWE-89: SQL Injection
 - CWE-79: Cross-site Scripting (XSS)
 - CWE-78: OS Command Injection
 - CWE-22: Path Traversal
 - CWE-798: Use of Hard-coded Credentials
 - CWE-327: Use of a Broken or Risky Cryptographic Algorithm
 - CWE-352: Cross-Site Request Forgery (CSRF)
-- this should ring a bell from Lab 4.8
 - CWE-502: Deserialization of Untrusted Data
 - CWE-200: Information Exposure

CWE top 25 most dangerous software weaknesses				
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False Positives

- False positives
 - Every SAST tool reports things that aren't necessarily vulnerabilities
 - Because
 - A SAST tool that finds nothing is broken
 - A SAST tool that finds everything is useless
 - The tool errs on the side of reporting, leaves judgment to humans
 - "False positive" means the tool was wrong about the data flow or the risk

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False Positives

- False positives
 - Every SAST tool produces them
 - This is a deliberate design choice
 - The tool defers judgment to a human (you) for the cases where it can't be certain
 - The skill is distinguishing real findings from false positives
 - Two general mistake types:
 - *Accept everything*: treat every finding as real, fix or suppress all of them, burn out on the volume
 - *Dismiss everything*: assume the tool doesn't understand and dismiss findings without analysis, miss real issues

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Triage Workflow

- Confirm
 - Read the data flow trace the tool provides
 - Most SAST tools show the path from source to sink, line by line.
 - Walk through it mentally
 - Is the data actually untrusted?
 - Does it actually reach the sink?
 - Is there a sanitizer the tool missed?
 - At the end of this step, you have a position
 - Yes, this is real
 - Or no, the tool got something wrong.

For each finding:

1. CONFIRM -- Read the data flow. Does the alleged vulnerability exist?
2. CLASSIFY -- Pick one:
 - True positive, fix now
 - True positive, accepted risk (document)
 - False positive (document why)
 - Out of scope (test code, generated, etc.)
3. ACT -- Fix the code OR document the suppression OR confirm scope
4. VERIFY -- Re-scan to confirm fix or that suppression applied correctly

Triage Workflow

- Classify
 - Four categories cover almost every finding
 - True positive, fix now
 - The vulnerability is real, has no compensating control, should be remediated immediately
 - SQL injection in user-facing code is the canonical example.
 - True positive, accepted risk
 - The vulnerability is real but mitigated by other factors
 - The endpoint is internal-only and behind authentication
 - The system is air-gapped
 - The compensating control is documented and reviewed by security
 - The classification acknowledges the issue while explaining why immediate fix isn't required.

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Triage Workflow

- Classify
 - False positive: the tool is wrong
 - Either the data flow doesn't actually exist (sanitizer missed)
 - Or the alleged risk doesn't apply (the sink isn't actually sensitive)
 - Document why so a future developer or auditor needs to understand the reasoning
 - Out of scope:
 - The finding is in code that doesn't get deployed
 - Test code, generated code, third-party libraries
 - Often these can be excluded by configuration before scanning, but if they sneak in, classify them as out of scope with a note.

For each finding:

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Triage Workflow

- Act
 - For "fix now," change the code
 - For "accepted risk" and "false positive," create the suppression in the tool with the documented justification
 - For "out of scope," either fix the scan configuration or mark them in the tool.
- Verify
 - Always re-scan after fixing
 - The finding should be gone (or at least transformed into a different finding).
 - If it's still there, your fix didn't address what the tool was complaining about.

For each finding:

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Example

- Triage
 - Read the trace
 - Does the user-supplied accountId actually reach the SQL execution unsanitized?
 - Inspect the code.
 - If the query uses + concatenation: it's real
 - If it uses ? placeholders with `setString: false` positive, then the tool missed the parameterization
 - Decide
 - True positive, fix now
 - False positive, suppress with justification.
 - Re-scan
 - Confirm the finding is gone after the fix

[HIGH] CWE-89 SQL Injection in TransactionRepository.java:42

Source: request.getParameter("accountId") at TransactionController.java:18

Path: accountId → transactionService.getByAccount(accountId) → ...

Sink: statement.executeQuery(query) at TransactionRepository.java:42

Suggested fix: Use parameterized queries.

Severity Levels and Prioritization

- Not all findings are equal
 - Critical / High
 - Direct exploitability, business impact
 - RCE, SQL injection, hardcoded prod credentials
 - Medium
 - Exploitable in some contexts
 - XSS in admin pages, weak crypto for non-secret data
 - Low
 - Defense-in-depth issues
 - Verbose error messages, deprecated APIs, missing security headers
 - Info
 - Code quality observations, not exploitable on their own
 - CI pipeline typically blocks merge on Critical/High; tracks Medium; informational for Low

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Path: accountId → transactionService.getByAccount(accountId) → ...

Sink: statement.executeQuery(query) at TransactionRepository.java:42

Suggested fix: Use parameterized queries.

Mapping SonarQube to Checkmarx

- Quick comparison

- Cost

- SonarQube Community is free and open source
 - Checkmarx is a commercial product with per-seat or per-scan licensing

- Deployment

- SonarQube Community runs locally in a Docker container
 - Checkmarx is typically a centrally-managed enterprise installation or SaaS subscription

- Setup time for the labs

- SonarQube takes minutes (one Docker command)
 - Checkmarx requires procurement, account provisioning, and security team coordination over weeks

Concept	SonarQube (lab)	Checkmarx (likely at the bank)
Severity	Blocker / Critical / Major / Minor / Info	Critical / High / Medium / Low / Information
Finding identity	Issue with key, rule, severity, location	Result with query name, severity, CWE
Triage status	Open / Confirmed / Resolved / Reopened	To Verify / Confirmed / Not Exploitable / Proposed Not Exploitable
False positive flag	Resolution: "False Positive" with comment	Mark as "Not Exploitable" with justification
Vulnerability vocab	CWE numbers	CWE numbers (same)
Scan trigger	Maven/Gradle plugin or CLI	CI integration, IDE plugin, CLI

Mapping SonarQube to Checkmarx

- Quick comparison
 - Java coverage
 - Both produce strong findings on Java code
 - The specific findings differ but the categories overlap substantially
 - Vulnerability vocabulary
 - Both use CWE numbers, so a finding's classification is portable between tools
 - Triage workflow
 - Both support the same workflow only the UI vocabulary differs
 - Banking adoption
 - Checkmarx is more common in large banks for the central CI pipeline
 - SonarQube is more common at the team/project level and for code quality analysis alongside security

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Suppression Justifications

- Writing Justifications for audit
 - Bad:
 - False positive."
 - "I don't think this is a problem."
 - Better:
 - "False positive. The tool identified accountId flowing to executeQuery without sanitization, but inspection shows the JDBC PreparedStatement uses positional parameters (line 42, statement.setString(1, accountId)).
 - The tool's data flow analysis did not recognize the parameterization through the preparedStatement / setString idiom. No remediation required."
 - Best:
 - [the better text above] +
 - "Reviewed with [security team contact] on [date]. Filed as suppression #1234 with reference to internal AppSec wiki page <https://...>"

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Suppression Justifications

- A bad justification ("false positive")
 - Tells the future reader nothing.
 - They have to redo your analysis from scratch, often without the context you had
 - They may incorrectly accept your dismissal
 - They may incorrectly reject it
 - Either way, you've left them worse off than necessary.

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Suppression Justifications

- A good justification has three properties:
 - States the tool's allegation
 - "The tool identified X flowing to Y."
 - This shows you understood what the tool was reporting, not just that you disagreed.
 - Explains the actual code behavior
 - "Inspection shows the code uses parameterized queries via PreparedStatement.setString."
 - This is the technical fact that contradicts the tool's allegation.
 - Identifies WHY the tool was wrong
 - "The tool's data flow analysis did not recognize the JDBC parameterization idiom."
 - This is the diagnostic insight that explains the false positive at the level of the tool's behavior, not just the code.

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Suppression Justifications

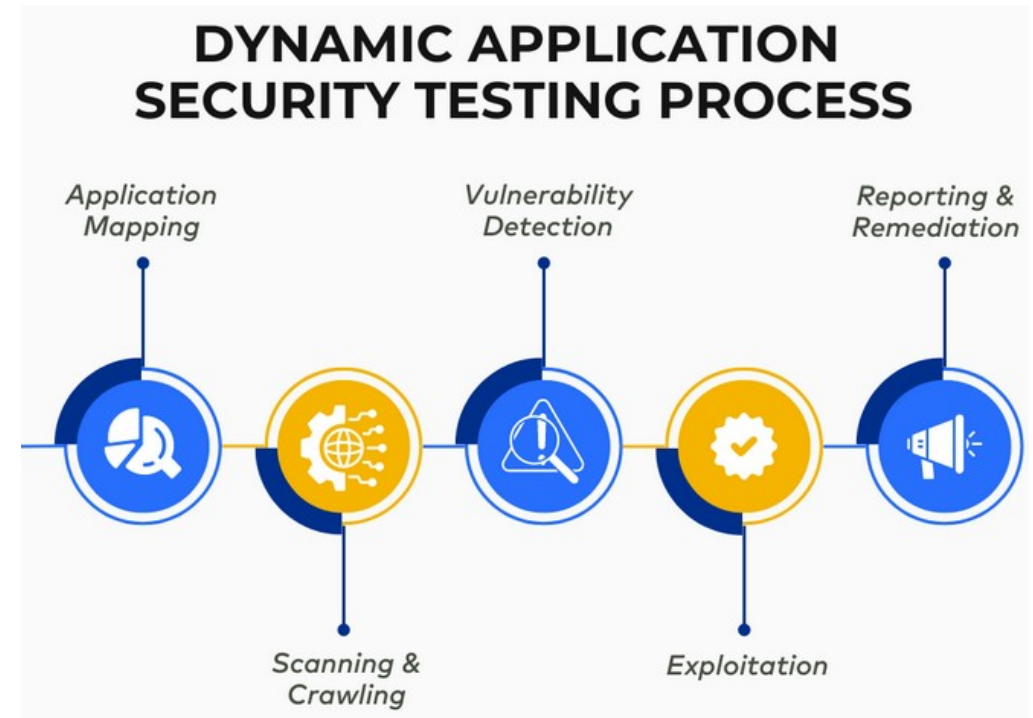
- The "Best" version
 - Adds review and traceability
 - A senior security person reviewed and concurred
 - A unique suppression ID in the tracking system links to the conversation
 - A wiki link or document reference provides extended context
 - This level of rigor is appropriate for high-severity findings or for codebases under regulatory scrutiny

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Dynamic Security Application Testing

DAST

- Dynamic application security testing
 - Dynamic
 - Runs against the actual running application
 - Black-box
 - The scanner does not see the source code, only the application's externally observable behavior
 - Late in the SDLC
 - Typically runs against staging or pre-production environments
 - The application is started
 - The scanner crawls it and discovers endpoints
 - Then probes those endpoints with crafted attack payloads
 - Observes responses to determine whether the application is vulnerable



DAST Scan Lifecycle

- Lifecycle

- Deploy

- DAST requires a running application
 - Biggest practical constraint
 - Unlike SAST, DAST needs an environment to point at
 - Usually this is a staging environment, a deployed copy of the application that mirrors production.

- Crawl

- The scanner discovers what URLs the application has.
 - For traditional server-rendered apps with HTML links, this is straightforward
 - Follow every `<a href>`, submit every `<form>`
 - For modern SPAs like React crawling is harder because URLs are constructed by JavaScript at runtime.
 - Some DAST tools include a JavaScript engine to handle this
 - Others rely on developers providing a list of URLs to scan, often via a Postman/OpenAPI exports

- | | |
|------------|--|
| 1. DEPLOY | -- Application is running in staging environment (real database, real config, real network) |
| 2. CRAWL | -- Scanner discovers endpoints by following links, submitting forms, exploring the URL space |
| 3. PROBE | -- For each endpoint, send crafted attack payloads: <ul style="list-style-type: none">- "' OR 1=1--" for SQL injection- "<script>alert(1)</script>" for XSS- "../..../etc/passwd" for path traversal |
| 4. OBSERVE | -- Watch responses for signs the attack succeeded: <ul style="list-style-type: none">- SQL errors in response- Reflected payloads in HTML- File contents leaking through |
| 5. REPORT | -- Findings include the URL, the payload, the response, and a CWE classification |

DAST Scan Lifecycle

- Lifecycle

- Probe

- The scanner has a library of attack payloads tagged by vulnerability type
 - For each form field, query parameter, or header, it tries the relevant payloads
 - This is genuinely sending attack traffic to the application.

- Observe

- The scanner looks for signals that an attack worked
 - SQL injection signals: database error messages in the response.
 - XSS signals: the payload appearing in the response unescaped
 - Path traversal signals: file contents in the response
 - Each signal corresponds to a different finding type.

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DAST Scan Lifecycle

- Lifecycle

- Report

- Findings include the URL, the payload that triggered the finding, the response, and a CWE classification
 - Same CWE vocabulary as SAST.

- DAST scans are slow

- A thorough DAST scan against a moderately-sized web application can take hours
 - They also generate noise in production logs
 - This is why DAST runs in staging, not production.

- | | |
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DAST and SAST

- What DAST does that SAST doesn't

- Runtime configuration issues

- Environment variables
 - Deployed `application.yml` values
 - Infrastructure settings

- Deployment misconfigurations

- Exposed admin endpoints
 - Debug mode enabled
 - Default credentials

- Authentication flow problems

- Session fixation
 - Token replay
 - password reset bypasses

- | | |
|------------|--|
| 1. DEPLOY | -- Application is running in staging environment (real database, real config, real network) |
| 2. CRAWL | -- Scanner discovers endpoints by following links, submitting forms, exploring the URL space |
| 3. PROBE | -- For each endpoint, send crafted attack payloads: <ul style="list-style-type: none">- "' OR 1=1--" for SQL injection- "<script>alert(1)</script>" for XSS- "../..../etc/passwd" for path traversal |
| 4. OBSERVE | -- Watch responses for signs the attack succeeded: <ul style="list-style-type: none">- SQL errors in response- Reflected payloads in HTML- File contents leaking through |
| 5. REPORT | -- Findings include the URL, the payload, the response, and a CWE classification |

DAST and SAST

- What DAST does that SAST doesn't
 - Server-level vulnerabilities
 - Web server misconfigurations
 - Missing security headers
 - Weak TLS
 - Real-world exploitability
 - The SAST tool says "this could be exploited";
 - DAST proves "this WAS exploited"

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DAST and SAST

- What SAST does that DAST doesn't

- Code paths the scanner cannot reach

- Internal APIs
 - Admin endpoints
 - Code behind feature flags
 - Error paths

- Code quality issues that aren't yet exploitable

- Weak crypto in development
 - Hardcoded credentials in source
 - Deprecated APIs

- | | |
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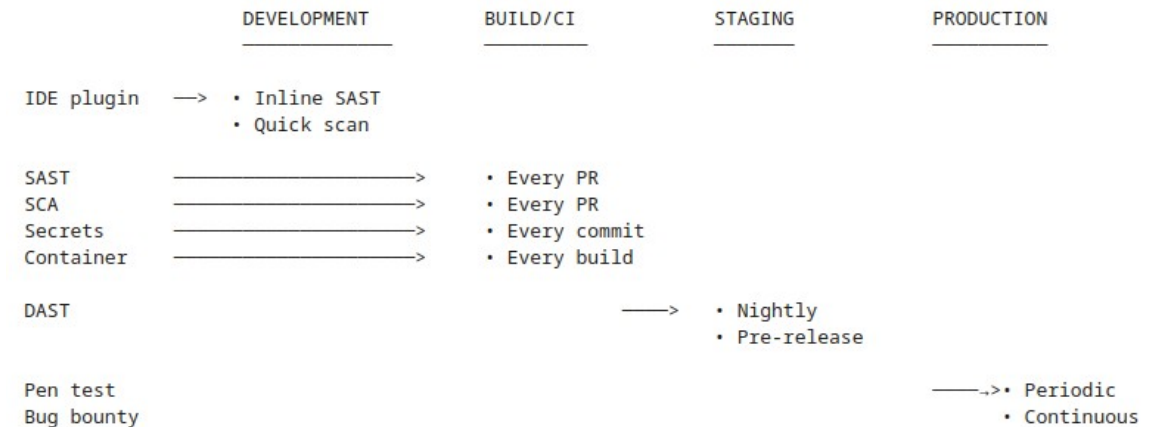
DAST and SAST

- What SAST does that DAST doesn't
 - Vulnerabilities in
 - Non-exposed code
 - Internal microservice APIs
 - Batch jobs and scheduled tasks
 - Per-finding precision
 - SAST points at the exact line
 - DAST points at the URL
 - Coverage of cold paths
 - Code that runs rarely like error handlers, infrequent business cases
 - Gets analyzed by SAST regardless of whether it runs or not

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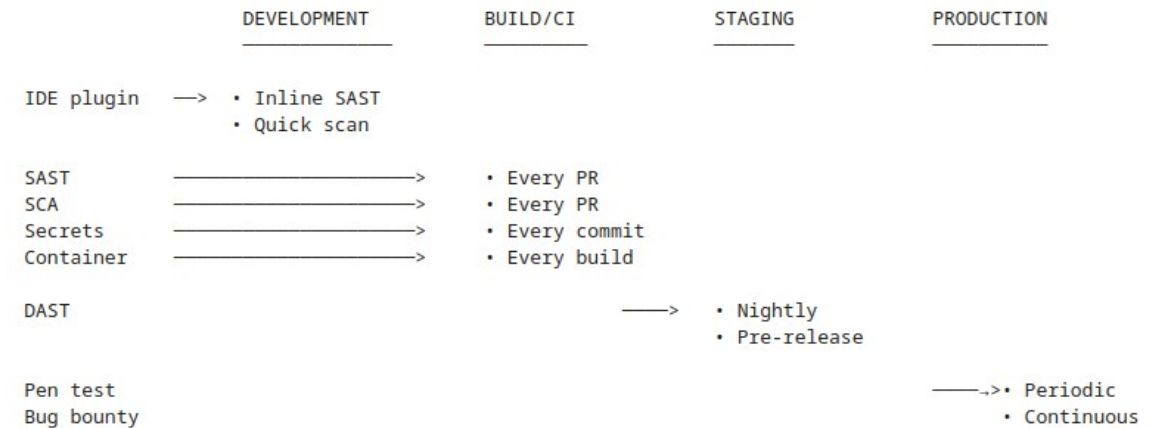
SDLC Fit

- For the bank context
 - All these layers exist in mature programs.
- Banks typically have
 - SAST and SCA in CI
 - For the bankbff, this would be triggered on every push
 - DAST runs against staging environments before production releases
 - Penetration tests at least annually, often more often for high-value applications
 - A bug bounty or vulnerability disclosure program for external reporting



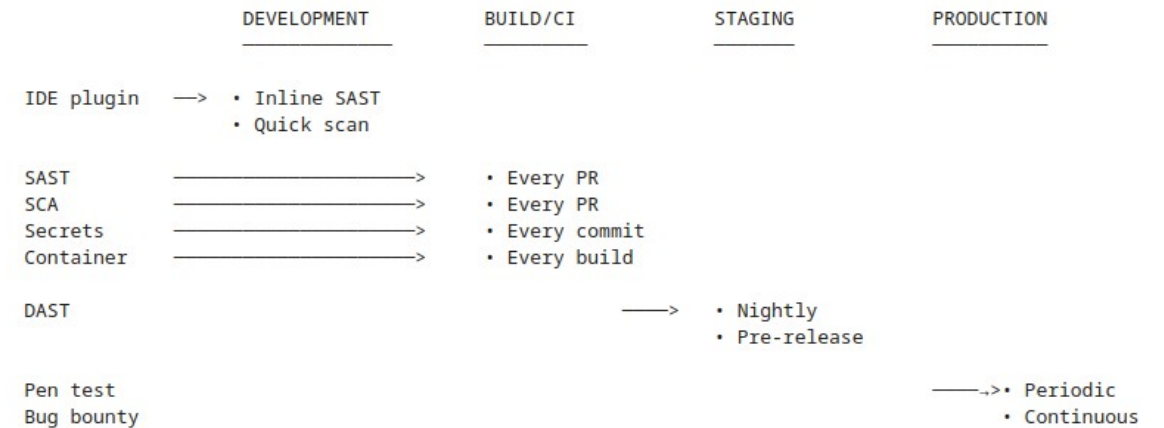
Other AppSec Layers

- Beyond SAST and DAST
 - SCA (Software Composition Analysis)
 - Scan dependencies for known vulnerabilities (CVEs)
 - Secrets scanning
 - Detect API keys, passwords, certificates accidentally committed to source control
 - Container scanning
 - Analyze Docker images for vulnerable OS packages and configurations
 - IAST (Interactive Application Security Testing)
 - Runtime instrumentation that combines SAST and DAST signals
 - Penetration testing
 - Humans, manually, attacking the application to find what scanners cannot
 - Bug bounty
 - External researchers paid to find and report vulnerabilities



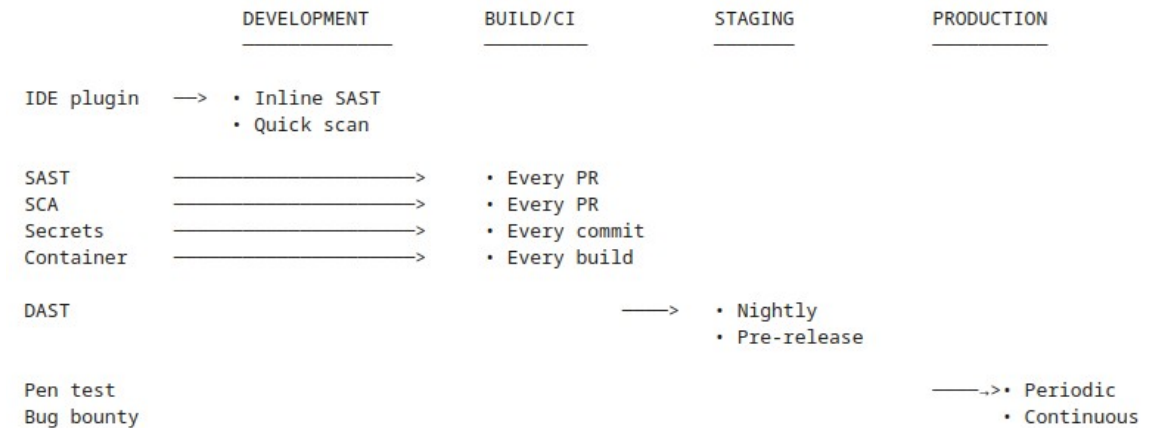
The Developer's Role

- You own
 - SAST findings
 - SCA findings
 - Secrets scanning findings
 - Fixing the vulnerabilities those tools surface
- You partially own
 - Unit and integration tests
 - Tests that catch security regressions



The Developer's Role

- Security engineers own
 - Running DAST scans
 - Configuring tooling
 - Maintaining baselines
 - Escalating findings
- Pen testers own
 - Finding what tools cannot find; their reports become work items for developers
- Bug bounty support own
 - Validating external reports
 - Valid findings become work items for developers



Questions?

